

Accuracy Limits of Causal Trees for Individualized Treatment Effects

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Abstract

Recursive decision trees are widely used to estimate heterogeneous causal treatment effects in experimental and observational studies. These methods are typically implemented using CART-type recursive partitioning, with splitting criteria designed to identify variation in treatment effects across covariate-defined subgroups. We study causal tree estimators based on adaptive recursive partitioning and establish lower bounds on their estimation accuracy. The class we analyze includes versions with and without sample splitting, based on common treatment effect and squared-error splitting criteria. Even in a constant-effect benchmark with randomized treatment assignment, causal trees constructed via standard CART-type splitting rules can have uniform-norm errors that decrease more slowly than any power of the sample size. The underlying mechanism is that greedy recursive partitioning selects highly imbalanced splits with nonvanishing probability, producing terminal nodes containing very few observations and leading to large estimation variance. We further show that sample splitting, often called “honesty,” does not remove this limitation. As a consequence, causal tree estimators may converge arbitrarily slowly uniformly over the covariate space. At the same time, these estimators can have small integrated mean squared error, showing that average accuracy can mask local inaccuracy. Our results also clarify the role of balanced partition assumptions in existing theoretical guarantees for causal forests and related ensemble methods.

Keywords: recursive partitioning, decision trees, causal inference, heterogeneous treatment effects

1 Introduction

Recursive decision trees have become a popular tool for estimating heterogeneous causal effects in both experimental and observational settings. These methods adapt the classical CART (Classification and Regression Tree) algorithm [Breiman et al., 1984] to causal inference by modifying the splitting criterion and, in some implementations, using sample splitting to separate tree construction from treatment effect estimation. Their simplicity and interpretability have made tree and recursive partitioning methods common in academic research and applied practice.

Examples include regression trunk methods and interaction tree methods for treatment-by-covariate interaction discovery [Dusseldorp and Meulman, 2004, Su et al., 2008, 2009]; recursive partitioning methods for treatment subgroup identification, including differential effect search [Lipkovich et al., 2011], virtual twin subgroup search [Foster et al., 2011], qualitative interaction trees [Dusseldorp and Van Mechelen, 2014], marginal tree methods [Kang et al., 2012], GUIDE treatment effect trees and model-based subgroup procedures [Loh et al., 2015, Seibold et al., 2016]; and the honest causal tree framework [Athey and Imbens, 2016]. Related extensions address instrumental variables [Bargagli Stoffi and Gnecco, 2020, Wang et al., 2022], survival or censored outcomes [Zhang et al., 2017], observational data with heterogeneity and fitness balancing [Zhang et al., 2018], causal interaction trees for observational data [Yang et al., 2022], and conditional inference trees [Venkatasubramaniam et al., 2022].

Despite their widespread use, the fundamental statistical properties of causal tree estimators and their associated inference procedures remain poorly understood for canonical CART-type implementations. Many existing analyses obtain positive guarantees by imposing balance or regularity conditions on the tree growing process. This paper studies heterogeneous treatment effect estimators based on adaptive recursive partitioning and establishes theoretical lower bounds on their estimation accuracy. The results show that standard CART-type recursive partitioning need not deliver uniform accuracy for heterogeneous treatment effect estimation.

Our main finding is that causal tree estimators constructed via standard CART-type greedy splitting can have uniform-norm errors that decrease more slowly than any power of n under basic conditions, where n denotes the sample size. This produces arbitrarily slow uniform convergence, even though average integrated error can remain small. Thus, good average performance can mask poor pointwise accuracy for individualized or subgroup-level decisions. This limitation arises even in the simplest setting in which the true treatment effect is constant over the feature space.

The mechanism behind this phenomenon is intrinsic to greedy recursive partitioning. CART-type splitting rules select highly imbalanced splits with nonvanishing probability, thereby generating terminal nodes with very few observations. Because treatment effects within each node are estimated by local averaging, these small cells induce large estimation variance and prevent the estimator from concentrating uniformly around the true conditional average treatment effect.

One way to address this small cell phenomenon is to regularize tree construction. In practice, this often takes the form of minimum node-size requirements. Such modifications may reduce finite-sample variance, but they also alter the class of feasible partitions of the procedure. When the conditional average treatment effect (CATE) is not locally constant, larger minimum node-size requirements can introduce approximation bias, and this tradeoff can be more pronounced under conditional heteroskedasticity. More aggressive regularization therefore changes the adaptivity that makes CART attractive. For this reason, in practice, causal tree methods are typically regularized through fixed or lightly tuned minimum node-size hyperparameters rather than through carefully tuned sample-size-dependent hyperparameters that grow with n , as would be required to address the kind of convergence issues raised here.

At present, however, the statistical behavior of CART-type estimators under such algorithmic regularization remains only partially understood in general settings where the partition is selected

adaptively from the data. For this reason, regularization alone does not provide a theoretically satisfactory resolution of the phenomenon we document unless one can formally characterize the resulting bias and variance tradeoff and establish valid inference for the resulting estimator. Moreover, even apart from these statistical concerns, it has been argued that imbalanced splits can be algorithmically beneficial for deeper trees because they preserve sample size after a bad split and allow the tree to recover downstream [Ishwaran, 2015].

Another frequently discussed modification of adaptive recursive partitioning methods is sample splitting, also known as *honesty*. By constructing the tree on one subsample and estimating treatment effects on an independent subsample, sample splitting separates partition selection from within leaf estimation and has been proposed as a way to reduce overfitting and facilitate inference [Athey and Imbens, 2016]. Honesty yields a modest improvement in the achievable convergence rate, but our results show that it does not remove the small cell mechanism behind the lower bounds. Even under honest sample splitting, causal tree estimators can have worst-case uniform errors that decrease more slowly than any power of n under basic conditions.

Our findings also have implications for theoretical guarantees established for quantile regression forests [Meinshausen, 2006], honest causal forests [Wager and Athey, 2018], generalized random forests [Athey et al., 2019], and random forest ensembles for Boolean interaction recovery [Behr et al., 2022]. Existing polynomial rate analyses of forest estimators often rely on conditions ensuring that each constituent tree generates approximately balanced partitions. In a common formulation, α -regularity requires each split to place at least an α fraction of the observations in the parent node into each child node. These assumptions facilitate theoretical analysis and define a more constrained class of tree-building rules than canonical CART-type greedy splitting, which can select highly imbalanced splits with nonvanishing probability. As a result, guarantees for causal forests that rely on approximately uniform partitions or α -regularity apply to a different algorithmic setting than standard CART-type splitting procedures. See Section 5 for further discussion.

From an inferential perspective, the same mechanism underlying our lower bounds also creates a challenge for subgroup-level inference based on causal trees and their ensembles. In regions where terminal nodes contain very few observations, the effective sample size may fail to increase with the overall sample size. In such settings, classical Gaussian approximations and standard error formulas need not be valid. Establishing distributional approximations for CART-based causal tree or forest estimators without imposing approximately uniform partition conditions therefore remains an open theoretical problem.

Before introducing the formal setup, it is useful to briefly summarize the intuition behind our results. Recursive decision trees select splits by optimizing a data-dependent criterion over many candidate partitions of the covariate space. When the underlying conditional expectation function is locally flat, as in the constant treatment effect model considered in this paper, the splitting criterion is driven primarily by stochastic fluctuations in the data. As a consequence, with nonvanishing probability the optimal split occurs near the boundary of a parent node, producing highly imbalanced child nodes. Because treatment effects within each node are estimated by local averaging, such splits generate regions with very few observations, leading to large estimation variance in those parts of the covariate space. For deeper trees, later refinements subdivide the imbalanced region rather than removing the initial small-cell problem, so at least one terminal descendant can retain enough of the fluctuation to prevent uniform convergence of the estimator.

1.1 Contributions and Related Literature

This paper makes three main contributions. First, we establish lower bounds on the uniform convergence rate of causal tree estimators constructed via CART-type recursive partitioning. These results show that such estimators can have uniform-norm errors that decrease more slowly than any power of n

under basic conditions, even in settings where the underlying treatment effect is constant. Second, we compare causal tree estimators with and without sample splitting. The lower bounds persist under honest sample splitting, which removes only a slowly varying $\sqrt{\log \log n}$ factor. Third, we clarify the implications of these findings for tree-based causal inference more broadly, including commonly used regularization strategies and theoretical guarantees for causal forests that rely on approximately uniform partitions or α -regularity conditions. Our analysis shows how these assumptions differ from canonical CART-type implementations and highlights the resulting challenges for valid inference based on tree-generated partitions.

Our work complements theoretical work on random forests [Scornet et al., 2015], recursive partitioning estimators [Chi et al., 2022, Klusowski and Tian, 2024, Mazumder and Wang, 2024], and partitioning-based regression [Cattaneo et al., 2024], and contributes to a growing body of formal limitations for tree-based methods. For example, Tan et al. [2022] demonstrated that regression trees are inefficient at estimating additive structure, regardless of the optimization strategy employed. Tan et al. [2024] established that the mixing time of Bayesian Additive Regression Trees (BART) [Chipman et al., 2010] can increase with the training sample size. Tan et al. [2026] showed that adaptive regression trees with Boolean covariates may require exponentially many samples in the dimension and can be inconsistent in high-dimensional settings. The Supplemental Appendix also develops companion lower bounds for standard CART regression trees. Those regression results clarify the relationship between our lower-bound framework and earlier large-sample analyses of CART decision stumps, including Bühlmann and Yu [2002] and Banerjee and McKeague [2007].

The proofs rely on new probabilistic arguments concerning the behavior of adaptive recursive partitioning procedures. The Supplemental Appendix develops nonasymptotic approximations for suprema of partial sums and related Gaussian processes, combining high-dimensional central limit theorems, Gaussian comparison inequalities, and strong approximation techniques from Chernozhukov et al. [2017], Chernozhukov et al. [2022], Csörgö and Révész [1981], Csörgö and Horváth [1997], El-Yaniv and Pechony [2009], Horváth [1993], Göing-Jaeschke and Yor [2003], Latała and Matlak [2017], Nazarov [2003], Petrov [2007], Shorack and Smythe [1976], Skorski [2023], and Zhdanov [2022]. As a technical byproduct of our analysis, we also identify and correct an error in Eicker [1979].

1.2 Organization

The remainder of the paper is organized as follows. Section 2 introduces the causal tree estimators and the data generating framework studied in the paper. Section 3 presents the assumptions underlying our theoretical analysis. Section 4 establishes lower bounds on the uniform convergence rate of causal tree estimators and provides complementary results on their integrated mean squared error. Section 5 discusses the implications of our results for recursive partitioning methods, causal forests, and inference procedures. Section 6 presents Monte Carlo and empirical resampling evidence illustrating the practical implications of the theoretical results. Section 7 concludes. Additional theoretical results, an overview of the proof strategy, and all technical proofs are reported in the Supplemental Appendix.

2 Setup

We introduce the class of causal tree estimators studied in the paper. These estimators combine three components (i) a within node estimator of the conditional average treatment effect (CATE), (ii) a recursive partitioning rule used to construct the tree, and (iii) a data usage scheme determining whether sample splitting is employed. Different choices of these components lead to the family of estimators analyzed in this paper.

The available data $\mathcal{D} = \{(y_i, d_i, \mathbf{x}_i^\top) : i = 1, 2, \dots, n\}$ is a random sample, where y_i is an outcome variable, $\mathbf{x}_i = (x_{i,1}, \dots, x_{i,p})^\top$ is a vector of pretreatment covariates, and d_i is a binary treatment

indicator. Employing standard potential outcomes notation [see, e.g., [Hernán and Robins, 2020](#)], we assume that $y_i = y_i(1)d_i + y_i(0)(1 - d_i)$, where $y_i(1)$ and $y_i(0)$ denote the potential outcomes under treatment and control, respectively. In experimental settings the treatment assignment is independent of both the potential outcomes and the covariates, that is, $(y_i(0), y_i(1), \mathbf{x}_i^\top) \perp\!\!\!\perp d_i$.

The parameter of interest is the conditional average treatment effect (CATE) function

$$\tau(\mathbf{x}) = \mathbb{E}[y_i(1) - y_i(0) \mid \mathbf{x}_i = \mathbf{x}],$$

which captures how treatment effects vary with observable pretreatment covariates. In experimental settings, the CATE function is identifiable because

$$\tau(\mathbf{x}) = \mathbb{E}\left[y_i \frac{d_i - \xi}{\xi(1 - \xi)} \mid \mathbf{x}_i = \mathbf{x}\right] = \mathbb{E}[y_i \mid d_i = 1, \mathbf{x}_i = \mathbf{x}] - \mathbb{E}[y_i \mid d_i = 0, \mathbf{x}_i = \mathbf{x}],$$

where the probability of treatment assignment $\xi = \mathbb{P}(d_i = 1)$ is known by virtue of the known randomization mechanism. The first equality represents the CATE as a single conditional expectation, $\mathbb{E}[\tilde{y}_i \mid \mathbf{x}_i = \mathbf{x}]$, where

$$\tilde{y}_i = y_i \frac{d_i - \xi}{\xi(1 - \xi)}$$

is the transformed outcome. The second equality expresses the CATE as the difference of two conditional expectation functions based on observed data.

Traditional semiparametric approaches estimate heterogeneous treatment effects by replacing these conditional expectations with nonparametric estimators. However, such methods may perform poorly in high-dimensional settings or when the structure of the regression functions (e.g., sparsity or additive separability) is unknown. Recursive partitioning offers a different strategy: it constructs covariate-defined subgroups and estimates treatment effects within the resulting terminal nodes.

Across the methods cited above, three components recur: a terminal node treatment effect estimator, a recursive partitioning rule that searches for treatment effect heterogeneity, and a data usage scheme governing whether the same observations are used for partition selection and effect estimation.

We study six causal tree estimators obtained by crossing three construction and estimation rules with two data usage schemes. The three rules are IPW-based variance maximization, DIM-based variance maximization, and SSE minimization; the two data usage schemes are no sample splitting and honest sample splitting. These choices cover the transformed outcome, treatment contrast, local model fit, and sample splitting designs used in causal tree implementations and related recursive partitioning procedures.

2.1 CATE Estimator

Leveraging these identification results, recursive partitioning methods construct terminal node treatment effects using pseudo outcomes [[Foster et al., 2011](#)], treated and control contrasts [[Su et al., 2009](#)], heterogeneity and fit criteria for observational data [[Zhang et al., 2018](#)], inverse-probability-weighted or doubly robust subgroup effect estimators [[Yang et al., 2022](#)], or local treatment effect models [[Loh et al., 2015](#), [Seibold et al., 2016](#)]. We analyze the following two CATE estimators based on a tree T and a dataset $\mathcal{D}_\tau$. Sections 2.2 and 2.3 discuss specific choices of T and $\mathcal{D}_\tau$, respectively. Let $\mathbb{1}(\cdot)$ be the indicator function.

Definition 1 (CATE Estimators). *Suppose T is the tree used, and $\mathcal{D}_\tau = \{(y_i, d_i, \mathbf{x}_i^\top) : i = 1, 2, \dots, n_\tau\}$, with $n_\tau \leq n$, is the dataset used. Let $\mathbf{t}$ be the unique terminal node in T containing $\mathbf{x} \in \mathcal{X}$.*

- The Inverse Probability Weighting (IPW) estimator is

$$\hat{\tau}_{\text{IPW}}(\mathbf{x}; \mathbf{T}, \mathcal{D}_\tau) = \frac{1}{n(\mathbf{t})} \sum_{i: \mathbf{x}_i \in \mathbf{t}} \tilde{y}_i,$$

where $n(\mathbf{t}) = \sum_{i=1}^{n_\tau} \mathbf{1}(\mathbf{x}_i \in \mathbf{t})$ is the “local” sample size. We set $\hat{\tau}_{\text{IPW}}(\mathbf{x}; \mathbf{T}, \mathcal{D}_\tau) = 0$ whenever $n(\mathbf{t}) = 0$.

- The Difference in Means (DIM) estimator is

$$\hat{\tau}_{\text{DIM}}(\mathbf{x}; \mathbf{T}, \mathcal{D}_\tau) = \frac{1}{n_1(\mathbf{t})} \sum_{i: \mathbf{x}_i \in \mathbf{t}} d_i y_i - \frac{1}{n_0(\mathbf{t})} \sum_{i: \mathbf{x}_i \in \mathbf{t}} (1 - d_i) y_i,$$

where $n_d(\mathbf{t}) = \sum_{i=1}^{n_\tau} \mathbf{1}(\mathbf{x}_i \in \mathbf{t}, d_i = d)$, for $d = 0, 1$, are the “local” sample sizes. We set $\hat{\tau}_{\text{DIM}}(\mathbf{x}; \mathbf{T}, \mathcal{D}_\tau) = 0$ whenever $n_0(\mathbf{t}) = 0$ or $n_1(\mathbf{t}) = 0$.

Both estimators, $\hat{\tau}_{\text{IPW}}(\mathbf{x}; \mathbf{T}, \mathcal{D}_\tau)$ and $\hat{\tau}_{\text{DIM}}(\mathbf{x}; \mathbf{T}, \mathcal{D}_\tau)$, rely on localization near $\mathbf{x}$. The tree construction $\mathbf{T}$ forms a partition of the support of the covariates $\mathcal{X}$, and estimation of $\tau(\mathbf{x})$ uses only observations with covariates $\mathbf{x}_i$ belonging to the cell in the partition covering $\mathbf{x} \in \mathcal{X}$. Therefore, given a tree (or partition), both estimators can be represented as nonparametric partitioning-based estimates of $\tau(\mathbf{x})$. See Györfi et al. [2002], Cattaneo et al. [2020], Cattaneo et al. [2026], and references therein.

Since the estimators $\hat{\tau}_{\text{IPW}}(\mathbf{x}; \mathbf{T}, \mathcal{D}_\tau)$ and $\hat{\tau}_{\text{DIM}}(\mathbf{x}; \mathbf{T}, \mathcal{D}_\tau)$ output a constant fit for all $\mathbf{x}$ within each terminal node of $\mathbf{T}$ (or cell in the partition), we define

$$\hat{\tau}_l(\mathbf{t}; \mathbf{T}, \mathcal{D}_\tau) = \hat{\tau}_l(\mathbf{x}; \mathbf{T}, \mathcal{D}_\tau), \quad l \in \{\text{IPW}, \text{DIM}\}, \quad \mathbf{x} \in \mathbf{t},$$

for all terminal nodes $\mathbf{t}$ of $\mathbf{T}$.

2.2 Tree Construction

An axis-aligned recursive decision tree is a predictive model that makes decisions by repeatedly splitting the data into subsets based on both outcome and covariate values. At each node, the algorithm selects the feature and threshold that best separate the data according to some criterion (e.g., squared error, Gini impurity, or entropy), and this process continues recursively until a stopping condition is met (e.g., maximum depth or pure terminal nodes). See Berk [2020], Zhang and Singer [2010], and references therein.

The most popular implementation of recursive decision trees is via the CART algorithm, which proceeds in a top-down, greedy manner through recursive binary splitting. Given a dataset $\mathcal{D}_\mathbf{T} = \{(y_i, d_i, \mathbf{x}_i^\top) : i = 1, 2, \dots, n_\mathbf{T}\}$, with $n_\mathbf{T} \leq n$, a parent node $\mathbf{t}$ in the tree (i.e., a region in $\mathcal{X}$) is divided into two child nodes, $\mathbf{t}_\mathbf{L}$ and $\mathbf{t}_\mathbf{R}$, by minimizing the sum of squares error (SSE),

$$\min_{1 \leq j \leq p} \min_{\beta_\mathbf{L}, \beta_\mathbf{R}, \varsigma \in \mathbb{R}} \sum_{\mathbf{x}_i \in \mathbf{t}} (y_i - \beta_\mathbf{L} \mathbf{1}(x_{ij} \leq \varsigma) - \beta_\mathbf{R} \mathbf{1}(x_{ij} > \varsigma))^2, \quad (1)$$

where the solution yields estimates $(\hat{\beta}_\mathbf{L}, \hat{\beta}_\mathbf{R}, \hat{\varsigma}, \hat{j})$ for the two child-node fitted values, the split point, and the split direction, respectively. Because splits occur along values of a single covariate, the induced partition of $\mathcal{X}$ is a collection of hyperrectangles. The resulting refinement of $\mathbf{t}$ produces child nodes $\mathbf{t}_\mathbf{L} = \{\mathbf{x} \in \mathbf{t} : \mathbf{e}_j^\top \mathbf{x} \leq \hat{\varsigma}\}$ and $\mathbf{t}_\mathbf{R} = \{\mathbf{x} \in \mathbf{t} : \mathbf{e}_j^\top \mathbf{x} > \hat{\varsigma}\}$.

The normal equations imply that $\hat{\beta}_\mathbf{L} = \frac{1}{n(\mathbf{t}_\mathbf{L})} \sum_{\mathbf{x}_i \in \mathbf{t}_\mathbf{L}} y_i$ and $\hat{\beta}_\mathbf{R} = \frac{1}{n(\mathbf{t}_\mathbf{R})} \sum_{\mathbf{x}_i \in \mathbf{t}_\mathbf{R}} y_i$, the respective sample means after splitting the parent node at $\mathbf{e}_j^\top \mathbf{x} = \hat{\varsigma}$. These child nodes become new parent nodes at the next level of the tree construction, and the procedure continues recursively until a desired depth

K is reached. A maximal decision tree of depth K iterates this construction until either (i) a node contains a single data point $(y_i, \mathbf{x}_i^\top)$ or (ii) all input values $\mathbf{x}_i$ and/or all response values y_i within the node are the same, although in general recursive tree constructions need not split every parent node.

Treatment effect trees adapt the CART algorithm by changing the split objective from overall outcome prediction to treatment effect heterogeneity, treatment-by-covariate interaction, or local treatment model fit. The methods cited above connect to the splitting criteria studied here in two main ways.

One line of work uses candidate splits to create child nodes with different estimated treatment effects. Interaction trees select splits using treatment-by-split interactions [Su et al., 2008, 2009], differential effect search uses subgroup treatment effect contrasts [Lipkovich et al., 2011], and qualitative interaction trees target qualitative treatment effect differences [Dusseldorp and Van Mechelen, 2014]. Causal trees [Athey and Imbens, 2016] and survival causal trees [Zhang et al., 2017] optimize treatment effect fit using estimated child-node effects. Instrumental variable tree and forest procedures [Bargagli Stoffi and Gnecco, 2020, Wang et al., 2022], observational data procedures that balance heterogeneity and fitness [Zhang et al., 2018], and causal interaction tree procedures for observational data [Yang et al., 2022] modify the child-node effect estimator, weighting scheme, or objective to accommodate the relevant assignment setting.

A second line of work grows trees by fitting local models that include treatment terms and then splitting to improve model fit or expose instability in treatment effect parameters. This includes regression trunk methods [Dusseldorp and Meulman, 2004], marginal trees for observational data [Kang et al., 2012], GUIDE treatment effect trees [Loh et al., 2015], and model-based subgroup procedures [Seibold et al., 2016]. Related pseudo-outcome and virtual twin procedures use generated treatment effect outcomes [Foster et al., 2011], and conditional inference treatment effect procedures use formal tests to decide where treatment effects vary [Venkatasubramaniam et al., 2022]. We focus on the following three CART-type criteria, which represent canonical split construction principles in the literature.

Definition 2 (Tree Construction). *Suppose $\mathcal{D}_T = \{(y_i, d_i, \mathbf{x}_i^\top) : i = 1, 2, \dots, n_T\}$, with $n_T \leq n$, is the dataset used to construct the tree T . There is a unique node $\mathbf{t}_0 = \mathcal{X}$ at initialization, and child nodes are generated by iterative axis-aligned splitting of the parent node based on either of the following two rules. Throughout this definition, a candidate split is valid only if both child nodes contain at least one construction-sample observation. When a split criterion uses treatment-arm means or node-specific treatment coefficients, the candidate is valid only if the corresponding treated and control denominators in both child nodes are positive; otherwise the candidate is omitted from the optimization. If a parent node has no valid split, it is left terminal. Ties are resolved by a fixed deterministic rule.*

- *Variance Maximization. A parent node $\mathbf{t}$ (i.e., a terminal node partitioning $\mathcal{X}$) in a previous tree T' is divided into two child nodes, $\mathbf{t}_L$ and $\mathbf{t}_R$, forming the new tree T , by maximizing*

$$\frac{n(\mathbf{t}_L)n(\mathbf{t}_R)}{n(\mathbf{t})} \left(\hat{\eta}_l(\mathbf{t}_L; T, \mathcal{D}_T) - \hat{\eta}_l(\mathbf{t}_R; T, \mathcal{D}_T) \right)^2, \quad l \in \{\text{IPW}, \text{DIM}\}. \quad (2)$$

The resulting causal trees are denoted by $T_{\text{IPW}}(\mathcal{D}_T)$ and $T_{\text{DIM}}(\mathcal{D}_T)$, respectively.

- *SSE Minimization. A parent node $\mathbf{t}$ (i.e., a terminal node partitioning $\mathcal{X}$) in the previous tree T' is divided into two child nodes, $\mathbf{t}_L$ and $\mathbf{t}_R$, forming the next tree T , by solving*

$$\min_{a_L, b_L, a_R, b_R \in \mathbb{R}} \sum_{\mathbf{x}_i \in \mathbf{t}_L} (y_i - a_L - b_L d_i)^2 + \sum_{\mathbf{x}_i \in \mathbf{t}_R} (y_i - a_R - b_R d_i)^2, \quad (3)$$

where only the data $\mathcal{D}_T$ is used. The resulting causal tree is denoted by $T_{\text{SSE}}(\mathcal{D}_T)$.

We use the variance maximization criterion in (2) to represent splitting rules based on treatment effect contrasts. It differs from the original CART criterion (1) in that it explicitly chooses the split with the largest weighted squared difference between the estimated treatment effects in the two child nodes. This criterion is closest to treatment-effect fit rules that evaluate candidate splits using estimated effects in the candidate child nodes. Interaction tree methods use treatment-by-split interactions [Su et al., 2008, 2009], differential effect search uses subgroup treatment effect contrasts [Lipkovich et al., 2011], and qualitative interaction procedures target qualitative subgroup treatment differences [Dusseldorp and Van Mechelen, 2014]. Instrumental variable tree and forest procedures [Bargagli Stoffi and Gnecco, 2020, Wang et al., 2022], observational data procedures that balance heterogeneity and fitness [Zhang et al., 2018], and causal interaction tree procedures for observational data [Yang et al., 2022] keep the recursive search over candidate splits, but modify the local effect estimator, weighting scheme, or objective for the relevant assignment setting.

For the IPW estimator, this rule is exactly equivalent to applying the CART criterion in (1) to the transformed outcome $\tilde{y}_i$. This transformed outcome satisfies $\mathbb{E}[\tilde{y}_i \mid \mathbf{x}_i = \mathbf{x}] = \tau(\mathbf{x})$ for all $\mathbf{x} \in \mathcal{X}$, and thus CART operates on an outcome whose conditional mean equals the CATE. The DIM estimator follows the same child-node contrast logic using treated and control differences in means.

We use SSE minimization to represent splitting rules based on local treatment model fit. For each candidate split, it fits separate regressions of the outcome on a treatment indicator in the two child nodes and chooses the split that most improves residual fit. This is closest to procedures that combine recursive partitioning with treatment-specific or node-specific regression models.

Regression trunk methods combine regression trees with treatment-covariate regression models [Dusseldorp and Meulman, 2004], marginal trees for observational data use likelihood-based regression trees to make treatment effects and propensities more homogeneous within leaves [Kang et al., 2012], and GUIDE treatment effect trees fit local models and use split tests to identify treatment effect variation [Loh et al., 2015]. Model-based subgroup procedures are related because they fit treatment models and split when treatment parameters are unstable across covariates [Seibold et al., 2016]. Conditional inference treatment effect methods use a more explicitly test-based implementation, so they are best viewed as related procedures for deciding where treatment effects vary rather than exact objective function matches [Venkatasubramaniam et al., 2022]. Supplemental Appendix, Section SA-3.3.1, derives the representation of (3) as a sum of treated and control variance gains.

Each of the causal recursive tree constructions leads to a distinct data-driven partition of $\mathcal{X}$. A key observation underlying our analysis is that these recursive procedures do not generate approximately uniform partitions of the covariate space, and thus known results in the nonparametric partitioning-based estimation literature [Györfi et al., 2002, Cattaneo et al., 2020, 2026] are not applicable. The Supplemental Appendix considers other recursive partitioning constructions, including the standard CART algorithm and variants thereof.

2.3 Sample Splitting

The final ingredient of the causal tree estimators concerns the data used at each stage of their construction. Sample splitting separates the data used to construct the partition from the data used to estimate treatment effects within leaves. In causal tree terminology, this design is often called “honesty” and has been proposed as a way to reduce overfitting and facilitate inference [Athey and Imbens, 2016]. Interaction tree methods [Su et al., 2008, 2009], differential effect search [Lipkovich et al., 2011], virtual twin subgroup search [Foster et al., 2011], and qualitative interaction trees [Dusseldorp and Van Mechelen, 2014] instead follow the broader CART tradition of selecting subgroups and estimating subgroup effects within one recursive partitioning workflow, with pruning, cross-validation, resampling, or testing-based rules used to control complexity and stability. We use no sample splitting and honesty as descriptive labels for these two data usage schemes.

To elucidate the relative merits of sample splitting, we consider two distinct scenarios (i) no sample splitting, where the same data is used throughout (as the original CART procedure is often implemented); and (ii) sample splitting, where two independent datasets are used, one for tree construction and the other for CATE estimation. Formally, we consider the following data usages and resulting treatment effect estimators.

Definition 3 (Sample Splitting and Estimators). *Recall Definition 1 and Definition 2, and that $\mathcal{D} = \{(y_i, d_i, \mathbf{x}_i^\top) : i = 1, 2, \dots, n\}$ is the available random sample.*

- *No Sample Splitting (NSS). The dataset $\mathcal{D}$ is used for both the tree construction and the treatment effect estimation, that is, $\mathcal{D}_\top = \mathcal{D}$ and $\mathcal{D}_\tau = \mathcal{D}$. The causal tree estimators are*

$$\begin{aligned}\hat{\tau}_{\text{IPW}}(\mathbf{x}) &= \hat{\tau}_{\text{IPW}}(\mathbf{x}; \mathbf{T}_{\text{IPW}}(\mathcal{D}), \mathcal{D}), \\ \hat{\tau}_{\text{DIM}}(\mathbf{x}) &= \hat{\tau}_{\text{DIM}}(\mathbf{x}; \mathbf{T}_{\text{DIM}}(\mathcal{D}), \mathcal{D}), \quad \text{and} \\ \hat{\tau}_{\text{SSE}}(\mathbf{x}) &= \hat{\tau}_{\text{DIM}}(\mathbf{x}; \mathbf{T}_{\text{SSE}}(\mathcal{D}), \mathcal{D}).\end{aligned}$$

- *Honesty (HON). The dataset $\mathcal{D}$ is divided into two independent datasets $\mathcal{D}_\top$ and $\mathcal{D}_\tau$ with sample sizes $n_\top$ and n_τ , respectively, and satisfying $n \lesssim n_\top, n_\tau \lesssim n$. The causal tree estimators are*

$$\begin{aligned}\check{\tau}_{\text{IPW}}(\mathbf{x}) &= \hat{\tau}_{\text{IPW}}(\mathbf{x}; \mathbf{T}_{\text{IPW}}(\mathcal{D}_\top), \mathcal{D}_\tau), \\ \check{\tau}_{\text{DIM}}(\mathbf{x}) &= \hat{\tau}_{\text{DIM}}(\mathbf{x}; \mathbf{T}_{\text{DIM}}(\mathcal{D}_\top), \mathcal{D}_\tau), \quad \text{and} \\ \check{\tau}_{\text{SSE}}(\mathbf{x}) &= \hat{\tau}_{\text{DIM}}(\mathbf{x}; \mathbf{T}_{\text{SSE}}(\mathcal{D}_\top), \mathcal{D}_\tau).\end{aligned}$$

The no sample splitting and sample splitting data usages are commonly encountered in the literature, and thus our results will speak directly to theoretical, methodological and empirical work relying on these designs. While the estimators $\hat{\tau}_l(\mathbf{x})$ and $\check{\tau}_l(\mathbf{x})$, $l \in \{\text{IPW}, \text{DIM}, \text{SSE}\}$, depend on the depth of the tree construction used, our notation keeps this dependence implicit except in the results whose conclusions require an explicit depth restriction. The integrated mean square bounds below keep the depth K explicit, and may therefore be read along sequences $K = K_n$.

3 Assumptions

The following assumption describes the data generating process used throughout the analysis.

Assumption 1 (Data Generating Process). *$\mathcal{D} = \{(y_i, d_i, \mathbf{x}_i^\top) : 1 \leq i \leq n\}$ is a random sample generated by i.i.d. latent vectors $(\mathbf{x}_i, d_i, \varepsilon_i(0), \varepsilon_i(1))$, where $y_i = d_i y_i(1) + (1 - d_i) y_i(0)$, $\mathbf{x}_i = (x_{i,1}, \dots, x_{i,p})^\top$, and the following conditions hold for all $d = 0, 1$ and $i = 1, 2, \dots, n$.*

- (i) $d_i \perp (\mathbf{x}_i, \varepsilon_i(0), \varepsilon_i(1))$, and $\xi = \mathbb{P}(d_i = 1) \in (0, 1)$.
- (ii) $y_i(d) = \mu_d(\mathbf{x}_i) + \varepsilon_i(d)$, with $\mathbb{E}[\varepsilon_i(d)] = 0$ and $\mathbf{x}_i \perp (\varepsilon_i(0), \varepsilon_i(1))$.
- (iii) $\mu_d(\mathbf{x}) = c_d$ for all $\mathbf{x} \in \mathcal{X}$, where c_d is some constant and $\mathcal{X}$ is the support of $\mathbf{x}_i$.
- (iv) $x_{i,1}, \dots, x_{i,p}$ are independent and continuously distributed.
- (v) There exists $\alpha > 0$ such that $\mathbb{E}[\exp(\lambda \varepsilon_i(d))] < \infty$ for all $|\lambda| < 1/\alpha$ and $\mathbb{E}[\varepsilon_i^2(d)] > 0$.

Assumption 1(i) corresponds to a simple randomized experiment with treatment probability $\xi \in (0, 1)$. Assumption 1(ii) specifies a canonical regression representation for the potential outcomes, where $\mu_d(\mathbf{x})$ denotes the conditional mean function and the potential-outcome error vector is jointly

independent of $\mathbf{x}_i$; arbitrary dependence between $\varepsilon_i(0)$ and $\varepsilon_i(1)$ is allowed unless stated otherwise. Assumption 1(iii) imposes a constant treatment effect, $\tau = c_1 - c_0$, across the covariate space. Assumption 1(iv) requires the covariates to be independent and continuously distributed. Because recursive decision trees are invariant to monotone transformations of the covariates, this condition can be replaced without loss of generality by assuming that $\mathbf{x}_i$ is uniformly distributed on $\mathcal{X} = [0, 1]^p$. Throughout the asymptotic results, p is fixed. Finally, Assumption 1(v) requires the potential outcome errors to be subexponential, or equivalently, to satisfy a Bernstein moment condition.

Because our goal is to establish lower bounds on the estimation accuracy of the causal tree estimators defined in Definition 3, it suffices to consider the constant treatment effect model in Assumption 1. The constant function belongs to all commonly studied smoothness classes, including Hölder classes and functions with bounded total variation. Consequently, lower bounds established under this model immediately extend to larger classes of data generating processes. Formally, for any estimator $\hat{\tau}(\mathbf{x})$ and any class of distributions $\mathcal{P}$ containing the distribution $\mathbb{P}_1$ satisfying Assumption 1,

$$\sup_{\mathbb{P} \in \mathcal{P}} \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{X}} |\hat{\tau}(\mathbf{x}) - \tau(\mathbf{x})| > \epsilon \right) \geq \mathbb{P}_1 \left(\sup_{\mathbf{x} \in \mathcal{X}} |\hat{\tau}(\mathbf{x}) - \tau(\mathbf{x})| > \epsilon \right),$$

for all $\epsilon > 0$, and for any data generating class $\mathcal{P}$ that includes the distribution $\mathbb{P}_1$ satisfying Assumption 1. In fact, the constant treatment effect model is a canonical case to consider in causal inference.

Assumption 1 also removes issues related to smoothing (or misspecification) bias, covariate-driven heteroskedasticity, and heavy-tailed errors. In particular, because the CATE function $\tau(\mathbf{x})$ is constant over $\mathcal{X}$, the results below are not driven by the usual boundary or smoothing bias arising in nonparametric estimation. The unbiasedness lemma in the Supplemental Appendix gives exact empty-cell bias expressions for all six estimators. For honest estimators, no residual symmetry condition is needed because the final estimation fold is independent of the selected partition. For no-sample-splitting IPW, the corresponding sufficient condition is central symmetry, conditional on the treatment assignments, of the transformed residuals $\tilde{\varepsilon}_i = \tilde{y}_i - \tau$; for no-sample-splitting DIM and SSE estimators, the corresponding sufficient condition is deterministic tie-breaking and joint central symmetry of the potential-outcome errors. These auxiliary symmetry conditions are not additional assumptions for the main lower-bound theorems; they are used only for the exact unbiasedness statements. Under these conditions, the estimators are exactly unbiased whenever the relevant terminal-node denominators are positive, and otherwise their bias is only the empty-cell convention bias described in the supplement.

Consequently, the lower bounds established below are not driven by bias. Instead, they arise because adaptive recursive tree constructions can generate highly imbalanced partitions with nonvanishing probability, producing terminal nodes that contain very few observations. These small cells lead to large estimation variance in some regions of $\mathcal{X}$, which ultimately prevents the estimator from achieving polynomial convergence rates.

Finally, the constant treatment effect model can also be interpreted as a local approximation to smooth heterogeneous treatment effect functions. Recursive partitioning estimators approximate $\tau(\mathbf{x})$ using piecewise constant functions over the tree partition, which corresponds to a Haar basis representation. Our results therefore extend to shrinking neighborhoods of smooth functions around the constant function when the signal to noise ratio is sufficiently small.

4 Main Results

The following theorem establishes our main lower bound on the uniform accuracy of causal tree estimators.

Theorem 1 (Uniform Accuracy). *Suppose Assumption 1 holds and the underlying causal tree has depth at most an integer $K \geq 1$, possibly depending on n , and fix $b \in (0, 1)$. Then the following conclusions hold for each $l \in \{\text{IPW}, \text{DIM}, \text{SSE}\}$.*

- *No Sample Splitting (NSS)*. Also assume that $2^K \log^2 n = o(n^{b/4} \sqrt{\log \log n})$ if $l \in \{\text{DIM}, \text{SSE}\}$. Then, there exist positive constants C_1 and C_2 such that

$$\liminf_{n \rightarrow \infty} \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{X}} |\hat{\tau}_l(\mathbf{x}) - \tau(\mathbf{x})| \geq C_1 n^{-b/2} \sqrt{\log \log n} \right) \geq C_2 b,$$

where C_1 and C_2 only depend on p and the distribution of $(y_i(0), y_i(1), d_i)$.

- *Honesty (HON)*. Also assume that $\rho \leq n_{\top}/n_{\tau} \leq \rho^{-1}$ for some $\rho \in (0, 1)$. Then, there exist positive constants C_3 and C_4 such that

$$\liminf_{n \rightarrow \infty} \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{X}} |\check{\tau}_l(\mathbf{x}) - \tau(\mathbf{x})| \geq C_3 n^{-b/2} \right) \geq C_4 b,$$

where C_3 and C_4 only depend on ρ , p , and the distribution of $(y_i(0), y_i(1), d_i)$.

The constants in Theorem 1 do not depend on K or n . Section SA-1 of the Supplemental Appendix maps Theorem 1 to the estimator-specific source results, and Section SA-1.2 gives the common proof strategy. The analysis studies suprema of partial sums and related Gaussian processes.

The estimator-specific source results in the supplement give sharper constants in several special cases, including a $1/e$ probability constant for the transformed-outcome IPW tree and for depth-one DIM. For SSE and for deeper NSS-DIM and NSS-SSE trees, additional transfer steps lead to generic positive constants. The rate condition in the NSS bullet is trivially satisfied for fixed- K tree constructions, and is not needed for NSS-IPW or HON procedures.

The theorem is established in a deliberately favorable benchmark: treatment assignment is randomized with known probability, the CATE is constant, errors have light tails, and the splitting rules are canonical CART-type greedy procedures without balance restrictions or minimum leaf-size requirements beyond nonempty child nodes. Thus, the lower bounds are not driven by treatment effect complexity, confounding, or smoothing bias.

Theorem 1 establishes lower bounds on the uniform convergence rate of the six causal tree estimators introduced in Section 2, with the stated depth condition for the deeper no sample splitting DIM and SSE trees. It therefore isolates the shared CART-type recursive partitioning mechanism across the estimator, splitting rule, and data usage choices considered here, rather than a feature of a single implementation. For procedures without sample splitting, the estimators $\hat{\tau}_{\text{IPW}}(\mathbf{x})$, $\hat{\tau}_{\text{DIM}}(\mathbf{x})$, and $\hat{\tau}_{\text{SSE}}(\mathbf{x})$ need not achieve a uniform convergence rate of order $n^{-b/2} \sqrt{\log \log n}$ for any $b > 0$ in the regimes covered by the theorem. In particular, these estimators converge more slowly than any polynomial rate in n , implying that their accuracy must deteriorate in some regions of the covariate space $\mathcal{X}$.

Sample splitting, often called “honesty,” decouples model selection from treatment effect estimation. The second result in Theorem 1 analyzes the corresponding honest causal tree estimators, $\check{\tau}_{\text{IPW}}(\mathbf{x})$, $\check{\tau}_{\text{DIM}}(\mathbf{x})$, and $\check{\tau}_{\text{SSE}}(\mathbf{x})$. The theorem shows that these estimators also need not achieve polynomial convergence rates. Sample splitting improves the attainable rate only by removing the slowly varying $\sqrt{\log \log n}$ factor.

Theorem 1 therefore identifies a limitation of adaptive decision tree methods when the goal is to estimate heterogeneous treatment effects uniformly over the covariate space. In contrast, the same estimators can achieve favorable estimation accuracy when their performance is measured on average over $\mathcal{X}$, as shown by the following result. Let $F_{\mathbf{X}}(\mathbf{x}) = \mathbb{P}(\mathbf{x}_i \leq \mathbf{x})$.

Theorem 2 (Integrated Mean Square Accuracy). *Suppose Assumption 1 holds and the underlying causal tree has depth at most an integer $K \geq 1$, possibly depending on n . Then the following conclusions hold for each $l \in \{\text{IPW}, \text{DIM}, \text{SSE}\}$.*

- *No Sample Splitting (NSS)*. There exists a positive constant C_1 such that

$$\mathbb{E} \left[\int_{\mathbf{x}} |\hat{\tau}_l(\mathbf{x}) - \tau(\mathbf{x})|^2 dF_{\mathbf{X}}(\mathbf{x}) \right] \leq C_1 \frac{2^K \log^4(n) \log(np)}{n},$$

where C_1 only depends on p and the distribution of $(y_i(0), y_i(1), d_i)$.

- *Honesty (HON)*. If $\rho \leq n_{\top}/n_{\tau} \leq \rho^{-1}$ for some $\rho \in (0, 1)$, then there exists a positive constant C_2 such that

$$\mathbb{E} \left[\int_{\mathbf{x}} |\tilde{\tau}_l(\mathbf{x}) - \tau(\mathbf{x})|^2 dF_{\mathbf{X}}(\mathbf{x}) \right] \leq C_2 \frac{2^K \log^5(n)}{n},$$

where C_2 only depends on ρ , p , and the distribution of $(y_i(0), y_i(1), d_i)$.

The constants in Theorem 2 do not depend on K or n , and the displayed bounds are explicit in K . Consequently, the bounds imply integrated mean square consistency along any sequence $K = K_n$ satisfying $2^{K_n} \log^4(n) \log(np)/n \rightarrow 0$ for no sample splitting and $2^{K_n} \log^5(n)/n \rightarrow 0$ for honest sample splitting.

The expectation bounds in Theorem 2 follow from the supplemental source results mapped at the opening of Supplemental Appendix, Section SA-1, and build on ideas from Györfi et al. [2002] and Klusowski and Tian [2024]. Importantly, the result applies only under Assumption 1, that is, when the CATE function is constant. The purpose of this theorem is to highlight that, even in the same setting where uniform convergence is slow, causal decision trees can still achieve favorable performance in an integrated mean squared sense. It remains an open question whether near-optimal mean square convergence rates can be achieved over larger classes of functions by adaptive decision trees constructed using CART-type procedures.

An interpretation of the contrast between Theorem 1 and Theorem 2 relates to the often discussed tension between causal inference and prediction in machine learning. Adaptive causal trees may perform poorly pointwise and still perform well on average over the covariate space. Thus, average accuracy does not imply reliable pointwise estimation. Methods that perform well on average can still be unreliable when the task requires accurate estimates at specific covariate values. Consequently, adaptive recursive partitioning should be used with caution for heterogeneous prediction or causal inference tasks where pointwise or subgroup-level accuracy is important.

From a technical perspective, the results in Theorem 2 are new in the context of causal tree estimation, particularly for the formal comparison between no sample splitting and honest implementations. Supplemental Appendix, Section SA-1.3, gives probability versions of these integrated error bounds under the same assumptions, with the same rates $2^K \log^4(n) \log(np)/n$ for no sample splitting and $2^K \log^5(n)/n$ for honest sample splitting under the stated sample-size balance condition.

5 Discussion

This section interprets the theoretical implications of Theorems 1 and 2 and related results for adaptive regression trees established in the Supplemental Appendix.

5.1 Decision Stumps

The generation of highly unbalanced cells in adaptive recursive partitioning has been recognized since the early development of CART and is often referred to as the *end cut preference*. Informally, when the signal is weak relative to sampling noise, the empirical splitting criterion may be optimized by thresholds located near the boundary of the parent node.

For example, in the context of standard CART regression without sample splitting, [Breiman et al. \[1984, Theorem 11.1\]](#) and [Ishwaran \[2015, Theorem 4\]](#) showed that in one dimension ($p = 1$), for each $\delta \in (0, 1)$, the first split has an end-cut preference: one child node contains at most a δ fraction of the sample with probability approaching one. If translated directly to CATE estimation, this result would rule out a uniform error bound at any fixed multiple of the nearly parametric scale $n^{-1/2}\sqrt{\log \log(n)}$, i.e., for any $C > 0$,

$$\liminf_{n \rightarrow \infty} \mathbb{P} \left(\sup_{x \in \mathcal{X}} |\hat{\tau}_l(x) - \tau(x)| \geq C \sigma n^{-1/2} \sqrt{\log \log(n)} \right) = 1.$$

In contrast, our results hold for every fixed $p \geq 1$ and precisely characterize the regions of the support $\mathcal{X}$ where the pointwise rates of estimation are slower than any polynomial in n ; see the decision-stump lower bounds for CART, IPW, DIM, and SSE in the Supplemental Appendix. Thus, existing theoretical results do not by themselves reveal the limitations of causal trees for pointwise estimation. Moreover, our analysis covers settings with sample splitting and shows that the lower bounds persist under this data usage design. Finally, our results apply to causal tree constructions that differ from, and are more complex than, standard CART regression trees.

5.2 Deeper Trees, Multivariate Covariates, and the Location of Small Cells

Our theoretical results show that, under [Assumption 1](#), the first split of an adaptive decision tree generates a small child cell with nonvanishing probability. When this child cell is later refined, its terminal descendants partition a region that already has limited mass. The lower-bound argument shows that at least one terminal descendant can retain enough of the initial fluctuation to force slow uniform convergence. This phenomenon becomes more pronounced as the covariate dimension increases ($p > 1$), precisely the regime in which tree-based methods are often employed to detect treatment effect heterogeneity.

Importantly, once earlier splits have isolated a parent node, later refinements can occur anywhere within that parent node and along any coordinate direction. As a consequence, adaptive tree constructions can generate anisotropic hyperrectangular cells with extremely small sample sizes in geometrically complex parts of the covariate space $\mathcal{X} \subseteq \mathbb{R}^p$, rather than only as simple one-dimensional end intervals.

5.3 Regularization and Bias

A natural response to the small cell phenomenon is to regularize the tree construction in order to prevent highly imbalanced splits. For instance, the algorithm may impose minimum node-size constraints or incorporate penalties designed to discourage overfitting. Such modifications can reduce estimation variance, but they also alter the class of admissible partitions.

Adaptive tree procedures can generate small cells because local refinement is useful for reducing approximation bias, or because sampling noise drives an imbalanced split. These two sources are difficult to separate in empirical applications. Our theoretical analysis isolates the variance mechanism by focusing on constant treatment effects. When treatment effects vary across the covariate space, regularization rules designed to eliminate small cells may introduce approximation bias and change overall convergence behavior. Supplemental Appendix, Section SA-1.4, gives additional discussion and a threshold example.

5.4 α -Regularity and Causal Random Forests

[Theorem 1](#) complements existing positive results by showing what can happen when the balance restrictions used in those analyses are removed. In particular, prior analyses of honest causal trees and forests [[Wager and Athey, 2018](#)] assume that each split allocates a fixed proportion of observations to

both child nodes, that is, $n(\mathbf{t}_L) \geq \alpha n(\mathbf{t})$ and $n(\mathbf{t}_R) \geq \alpha n(\mathbf{t})$, where $\alpha \in (0, 1/2]$. This condition, known as α -regularity, rules out the highly imbalanced splits that drive our lower bounds. Related balance restrictions also appear in quantile regression forests [Meinshausen, 2006], generalized random forests [Athey et al., 2019], and work on Boolean interaction recovery from random forest ensembles [Behr et al., 2022].

In particular, α -regularity may substantially alter the adaptive behavior of recursive partitioning. A balanced tree may require several successive splits to approximate a small subgroup that an unrestricted CART split could isolate immediately. Supplemental Appendix, Section SA-1.5, gives a threshold example illustrating this point.

For this reason, in practice, causal tree methods are often regularized through fixed or lightly tuned minimum node-size hyperparameters, rather than through carefully tuned sample-size-dependent hyperparameters that grow with n , as would be required to address the kind of convergence issues raised here; implementation examples are discussed in Supplemental Appendix, Section SA-1.4. Consequently, convergence guarantees derived under balance conditions apply to a more constrained algorithm than the CART-type procedures commonly used in empirical work. Extending such guarantees to canonical implementations would require additional regularization that modifies the estimator and introduces further bias and variance tradeoffs.

5.5 Implications for Inference

Theorem 1 also has direct implications for statistical inference based on adaptive causal trees. Because recursive partitioning generates highly imbalanced cells with nonvanishing probability, the effective sample size within some regions of the covariate space need not increase with the overall sample size. As a consequence, standard distributional approximations for (“honest”) causal tree estimators may fail to hold even after appropriate centering and scaling. In particular, Gaussian asymptotic approximations can break down in regions where terminal nodes contain only a small number of observations.

A related way to view the issue is through the weights that an estimator places on outcomes. Knaus [2024] studies such outcome weights for treatment effect estimators, including generalized random forests, and shows that implementation choices affect their properties.

This phenomenon can make commonly used inference procedures based on asymptotic normality unreliable. For example, confidence intervals of the form $\hat{\tau}_\ell(\mathbf{x}) \pm z_\alpha \cdot \text{Sd.Err.}(\hat{\tau}_\ell(\mathbf{x}))$ or $\check{\tau}_\ell(\mathbf{x}) \pm z_\alpha \cdot \text{Sd.Err.}(\check{\tau}_\ell(\mathbf{x}))$, where z_α denotes the usual quantile of the standard Gaussian distribution and $\text{Sd.Err.}(\cdot)$ is a standard error estimator, need not provide asymptotically valid coverage for $\tau(\mathbf{x})$ over many regions of $\mathcal{X} \subseteq \mathbb{R}^p$.

6 Numerical Evidence

6.1 Simulations

We first illustrate the implications of Theorem 1 in controlled designs. Figure 1 reports the pointwise root mean squared error, $\{\mathbb{E}[(\hat{\tau}_\ell(\mathbf{x}) - \tau)^2]\}^{1/2}$ and $\{\mathbb{E}[(\check{\tau}_\ell(\mathbf{x}) - \tau)^2]\}^{1/2}$, for $\ell \in \{\text{IPW}, \text{DIM}, \text{SSE}\}$ in a bivariate design ($p = 2$). The estimates are based on 2,000 Monte Carlo replications with $\tau = \mu_0 = \mu_1 = 0$, $\varepsilon_i(0), \varepsilon_i(1) \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(0, 1)$, $n = 1,000$, and $\mathbf{x}_i \stackrel{\text{i.i.d.}}{\sim} \text{Uniform}([0, 1]^2)$. For each of the six causal tree estimators shown in the figure, we vary the tree depth over $K \in \{1, \dots, 5\}$. The companion univariate design is reported in Supplemental Appendix, Section SA-1.7.

Two patterns are visible. First, for any fixed depth K , pointwise RMSE is smallest near the center of the covariate space and increases as $\mathbf{x}$ approaches the boundary. This pattern is the finite-sample counterpart of the small cell phenomenon established in Supplemental Appendix, Section SA-1.2. Boundary splits create highly imbalanced terminal nodes, thereby reducing the effective sample size

available for local averaging. Second, for any fixed evaluation point, RMSE increases with tree depth. Deeper trees create more terminal node boundaries, so a larger share of evaluation points is affected by the same local imbalance that drives the decision stump lower bound.

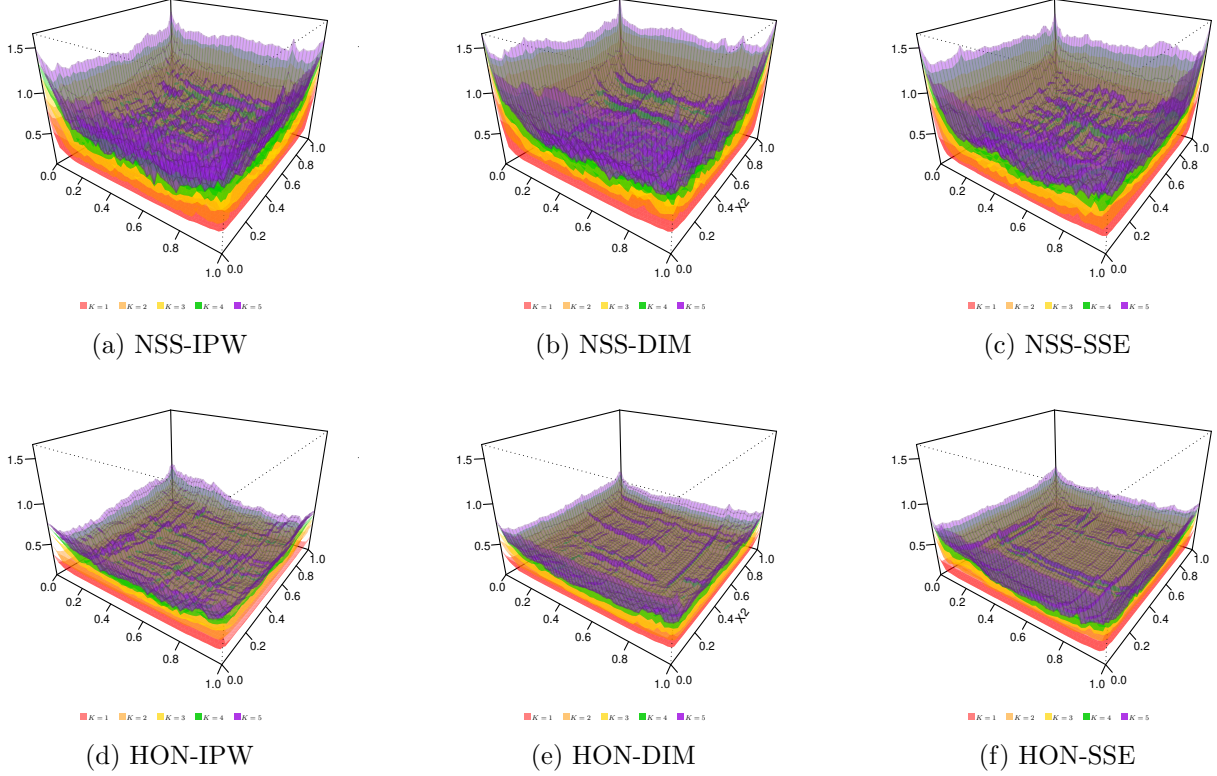


Figure 1: Synthetic Monte Carlo evidence for the bivariate design ($p = 2$). Each panel reports pointwise root mean squared error (RMSE) over the covariate support $[0, 1]^2$ for tree depths $K = 1, 2, \dots, 5$. Rows compare estimators without sample splitting (NSS) and with honest sample splitting (HON); columns compare inverse probability weighting (IPW), difference in means (DIM), and squared error (SSE) splitting criteria. RMSE is lowest near the center of the support and increases toward the boundary, where adaptive splitting is most likely to create small terminal cells. Results are based on 2,000 Monte Carlo replications.

6.2 Empirical JTPA Resampling

We next use an empirical resampling design based on the National Job Training Partnership Act (JTPA) Study. The JTPA study was a randomized evaluation of Title II-A employment and training services for economically disadvantaged adults and out-of-school youths in 16 local service delivery areas. The study design and main findings are reported in [Bloom et al. \[1997\]](#).

The resampling exercise preserves empirical features of the JTPA covariates but removes the original covariate-outcome association within each treatment arm by permuting covariates separately among treated and control units. This checks whether the small-cell mechanism remains visible under discrete mass points and uneven empirical support. Supplemental Appendix, Section SA-1.6, gives the preprocessing, thinning, and permutation details.

In Figure 2, RMSE again increases where the induced terminal nodes have limited empirical support. This occurs across splitting criteria and sample splitting schemes. The companion one-covariate design is reported in Supplemental Appendix, Section SA-1.7, and displays the same boundary pattern. The

sparse-support behavior seen in the Monte Carlo designs therefore also appears in the JTPA resampling exercise.

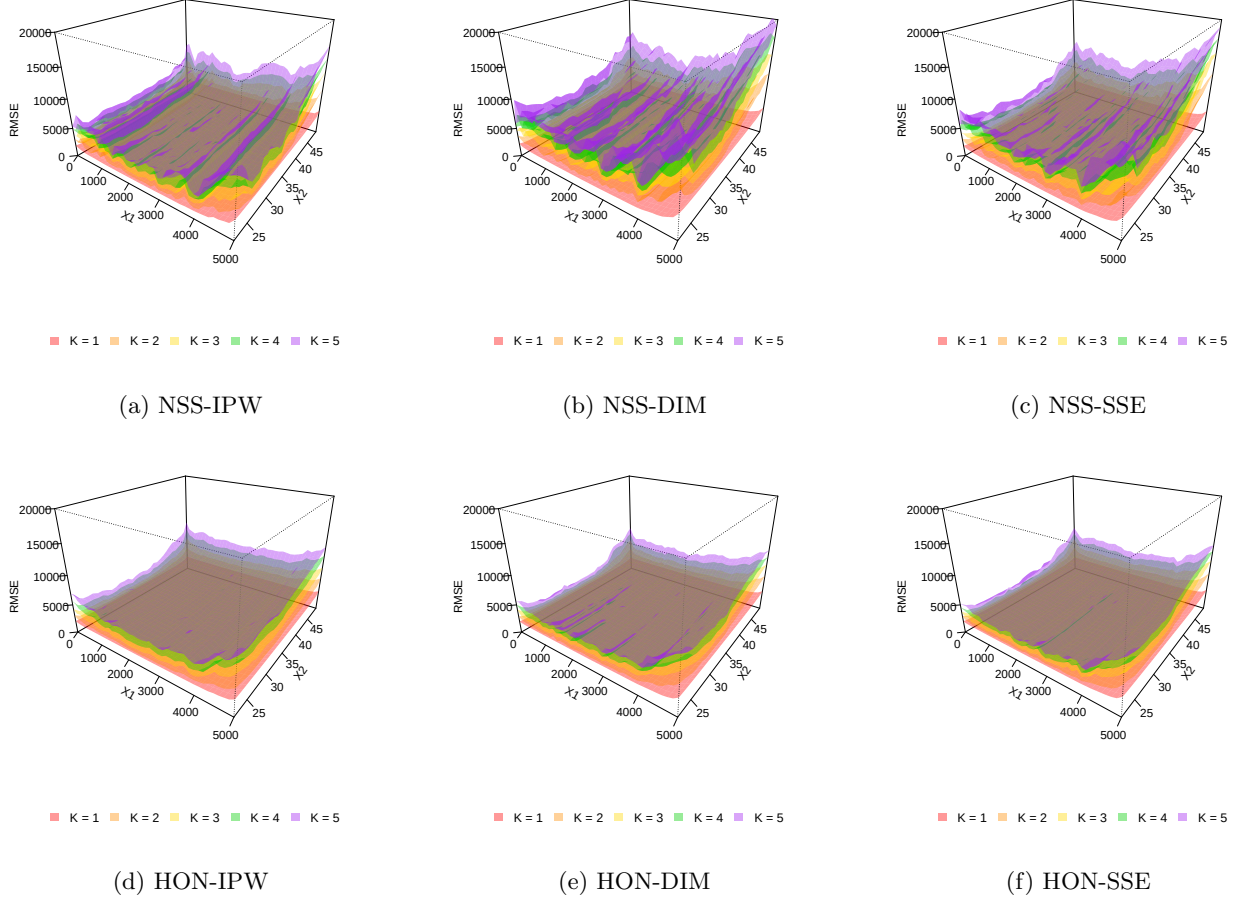


Figure 2: Empirical JTPA resampling evidence for the two-covariate design ($p = 2$). Panels report pointwise RMSE over the empirical joint covariate support after jointly permuting the covariate pair within treatment arms, for depths $K = 1, \dots, 5$. Rows compare NSS and HON; columns compare IPW, DIM, and SSE. RMSE rises in sparse regions of the support, matching the small-cell pattern in the simulations.

7 Conclusion

The results show that adaptive CART-type causal trees can have sharply different average and pointwise behavior. Even in a constant-effect benchmark, greedy splitting can create small terminal cells with nonvanishing probability. As a result, causal trees may have favorable integrated mean squared error while remaining unreliable at some covariate values, a distinction that matters for individualized decisions, subgroup conclusions, and inference.

Regularization is therefore central when causal trees are used for individualized treatment effect estimation or inference. Minimum node-size rules, balance restrictions, pruning, and related modifications can limit small cells, but they also change the estimator and introduce bias-variance tradeoffs and tuning-parameter choices. Existing positive theory often imposes balance or regularity conditions to obtain guarantees, rather than deriving them from canonical greedy CART splitting. Developing theory for practically regularized, adaptively selected causal tree partitions remains an important direction for future work.

The paper and Supplemental Appendix make this mechanism precise for transformed-outcome, difference-in-means, and squared-error splitting rules, with and without sample splitting. The proofs develop nonasymptotic approximations for adaptive split criteria, connect them to suprema of partial sums and Gaussian processes, transfer boundary split fluctuations through deeper trees, and provide companion regression-tree results and related technical corrections.

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